

**Report of BP-01 Major Overhauling
Work at WRPL VIRAMGAM
July-2020**

BP-01, Pump Bearing Temperature High Problem at WRPL

Viramgam

At WRPL Viramgam 5 nos of new booster's were commissioned under SMPL DBN project in 2017. Among these 5 nos, BP-01, 02 & 03 are of higher capacity (i.e. Rated Flow - 2100 KI/hr, Rated Head - 50 m with 605 KW rated power) and BP-04 & 05 are of lower capacity (i.e. Rated Flow -1325 KI/hr, Rated Head - 50 m with 380 KW rated power). The details of Booster Pumps are as below-

Booster Pump	Details
BP1, BP2 & BP3	Rated Flow - 2100 KI/hr, Rated Head - 50 m with 605 KW rated power, Make-BPCL
BP4 & BP5	Rated Flow -1325 KI/hr, Rated Head - 50 m with 380 KW rated power, Make-BPCL

BP-01 was not available for operation from 29.05.2020 due to pump bearing temperature high problem.

To attend the problem following action were taken.

- Pump shaft run out and alignment checked and found within limit.
- Pump rotation checked by hand, it was rotating freely.
- Pump bearing replacement done.

During further checking & trials, it is again observed that the pump bearing temperature is going high towards alarm/ trip set point. After many exercise at pump site, it had been decided to dismantle BP-01 for major overhaul.

Upon dismantling and inspection of pump followings were the observation.

- 2nos. of sleeve found in loose condition.
- Bottom end bush found damaged.

Following Maintenance Activities were Carried Out during overhauling:

- The rotor assembly was dismantled and inspection of all the parts was carried out.
- One number of spider bushes was found damaged. Boring of all three spider bushes up to marks disappeared.
- Checked pump shaft, transmission shaft, head shaft checked straightness for permissible limit. Found all shaft within permission limit.
- New sleeve manufactured by party and fitted interference fit.
- Muff coupling found ok.

- Impellers (02 no's) checked and found 1st stage impeller have minor hair crack in vein, welding was done and balancing of both impeller were carried out.
- Impellers wear ring and case wear rings was found ok within limit.
- Dynamic Balancing of all the rotary parts, UT of Shaft, Case Wear ring, Impeller Wear and Sleeve carried out.
- Fabrication of one Gun metal bush done and fitted at pump suction lower part (suction bell) with nitrogen cooling.
- After Repair of Pump, pump installation done and trail taken on 23.07.2020.

After overhauling pump its more than 110HRS as on 30.07.2020 (The details of major components of pump and all measurement details with photographs is attached as annexure-1).

Sunder Lal
OM, WRPL, Viramgam

SMNM, WRPL, Viramgam

Major Components of Pump and Measurement details are as follows:

Pump Shaft and Bowl:

- **Pump Shaft** The pump shaft is the central component of a centrifugal pump rotor and carries the impellers.
- **Bowl:** It houses the impeller and has metal to metal contact avoiding any deformation due to gaskets. Their number varies according to the head developed by the pump i.e. as per number of stages. It converts the kinetic energy of the liquid into pressure energy acting as diffuser and also serves as return channel to next stage impeller.

➤ **Pump Shaft and Bowl bush bore Before Maintenance Readings**

	Pump Shaft		Bowl Bush ID		Clearance	
	Aline	Across	Aline	Across	Aline	Across
1	80.00	79.99	Reading not taken due to bush ceased		Reading not taken due to bush ceased	
2	89.98	89.98	90.27	90.26	0.29	0.28
3	89.99	89.99	90.35	90.32	0.36	0.33
4	89.98	89.99	90.23	90.22	0.25	0.23

➤ **Pump Shaft and Bowl bush bore After Maintenance Readings**

	Pump Shaft		Bowl Bush ID		Clearance	
	Aline	Across	Aline	Across	Aline	Across
1	79.90	79.90	80.08	80.10	0.18	0.20
2	89.98	89.98	90.27	90.26	0.29	0.28
3	89.99	89.99	90.35	90.32	0.36	0.33
4	89.98	89.99	90.23	90.22	0.25	0.23

Transmission Shaft and Transmission column:

- **Transmission Shaft:** According to Depth of sump of barrel three transmission shafts are fitted in between pump shaft and head shaft which allows the pump to be lowered with respect to the mounting plane.
- **Transmission column:** According to the depth of the sump of barrel, three columns are fitted between discharge elbow and pump which allow the latter to be lowered with respect to the mounting plane.

➤ **Transmission Shaft sleeve and spider bush bore Before Maintenance Readings**

Transmission Shaft No.	Transmission Shaft sleeve OD		Spider Bush ID		Clearance	
	Aline	Across	Aline	Across	Aline	Across
1	99.81	99.82	99.99	99.97	0.18	0.15
2	99.80	99.80	99.99	99.99	0.19	0.19
3	99.80	99.80	99.96	99.97	0.16	0.17

➤ **Transmission Shaft sleeve and spider bush bore After Maintenance Readings**

Transmission Shaft No.	Transmission Shaft sleeve OD		Spider Bush ID		Clearance	
	Aline	Across	Aline	Across	Aline	Across
1	100.80	100.80	100.98	100.98	0.18	0.18
2	100.80	100.80	100.98	100.98	0.18	0.18
3	100.80	100.80	100.98	100.98	0.18	0.18

Head Shaft:

It connects motor shaft to the pump shaft/transmission shaft. Thrust bearing and mechanical seal sleeve is mounted on it.

Impeller:

The Impeller is single suction totally enclosed and dynamically balanced. The impellers have individual taper fit to accommodate impeller taper sleeve for mounting on the shaft or keyed to the shaft. In our pump impellers fitted on the shaft as a keyed to the shaft.

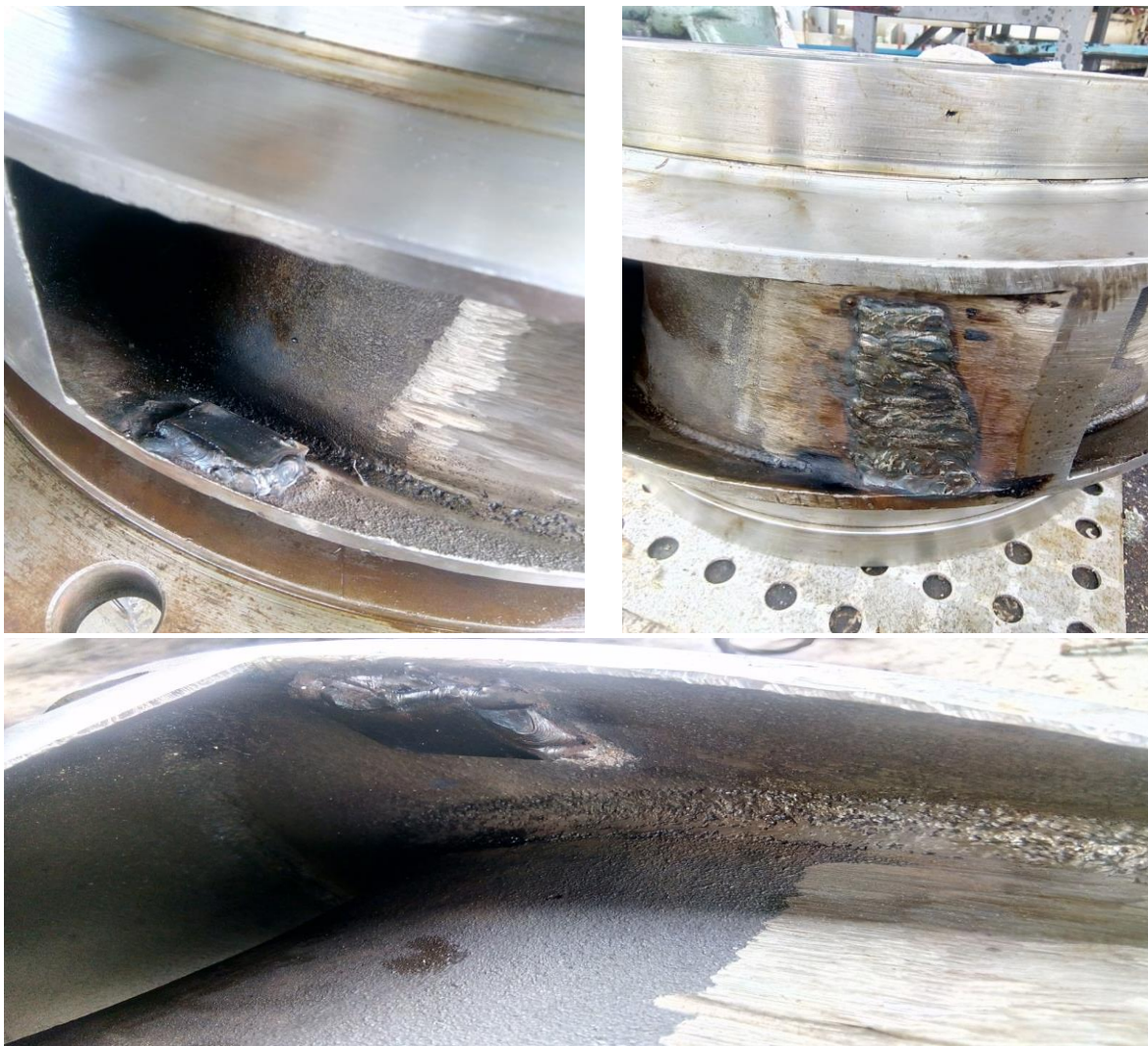
➤ **Impeller wear ring and case wear Readings are observed within limits.**

Impeller Stage	Ring	Impeller Wear Ring		Case Wear Ring		Clearance	
		Aline	Across	Aline	Across	Aline	Across
1st	Top	377.27	377.26	378.02	378.02	0.75	0.76
	Bottom	377.16	377.15	378.03	378.04	0.87	0.89
2nd	Top	377.33	377.33	378.04	378.06	0.71	0.73
	Bottom	377.31	377.33	378.04	378.03	0.73	0.70

Defects Found:-



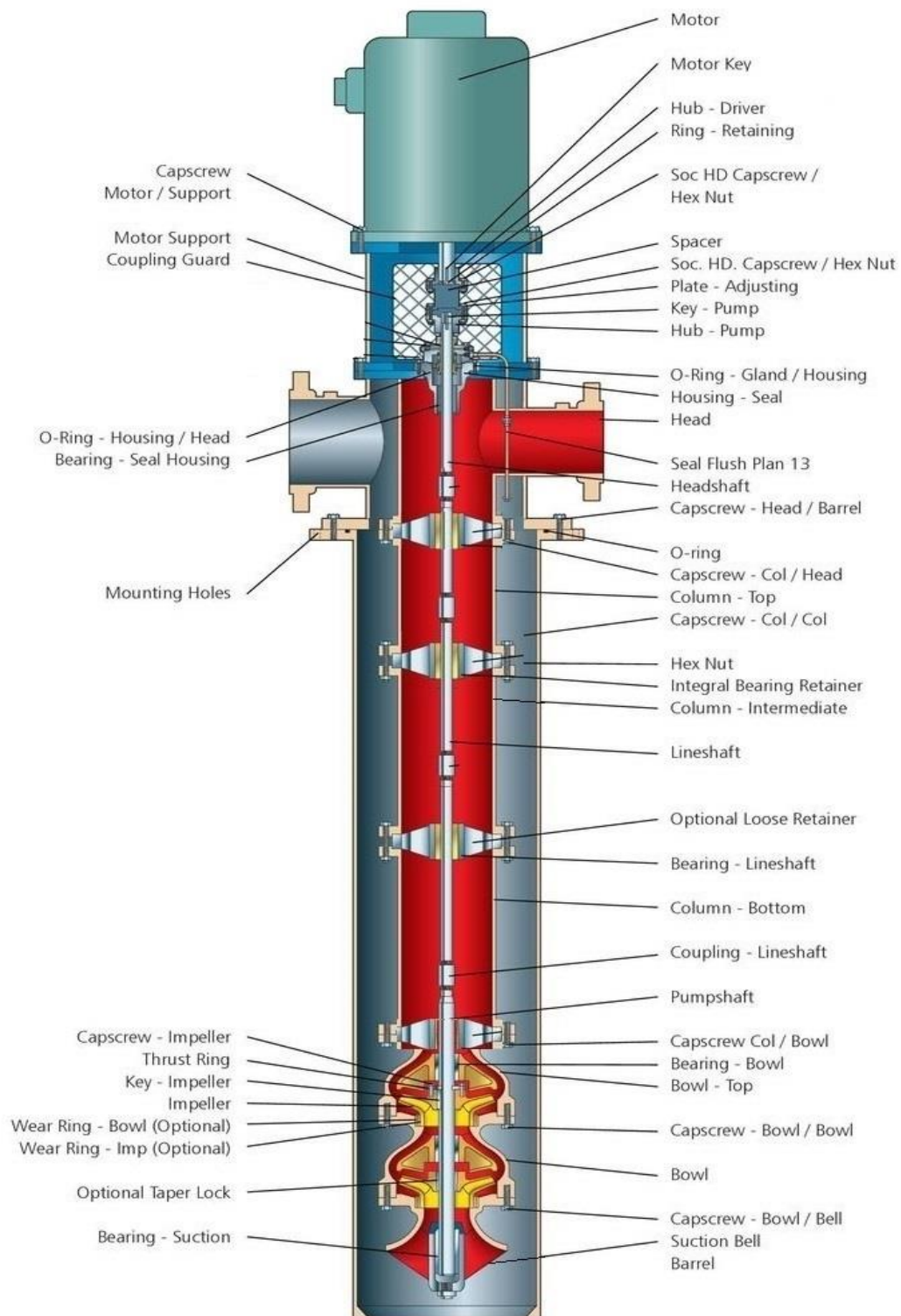
Suction bell bush found damage



Impeller crack in vane

Booster Pump reference Drawing:

Cross Sectional



Photographs of activity are as below-



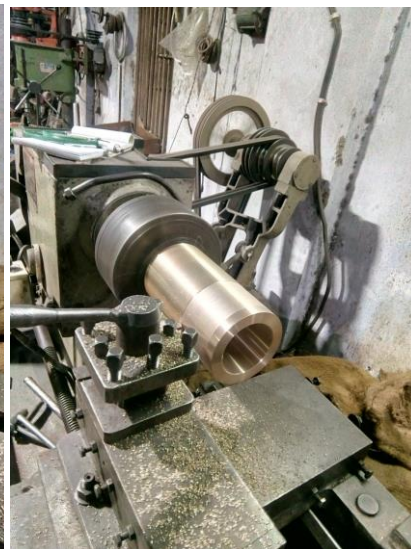
Shaft straightness checking on V-blocks.



Checking of Bearing Housing Sleeve.



Suction Bell machining work



Machining of Gun metal bush





Gun metal Bush shrink fitting process using Liquid Nitrogen into suction bell

Shrink Fitting

A cost efficient method of assembling and disassembling new and replacement fine tolerance components. An ideal solution for:

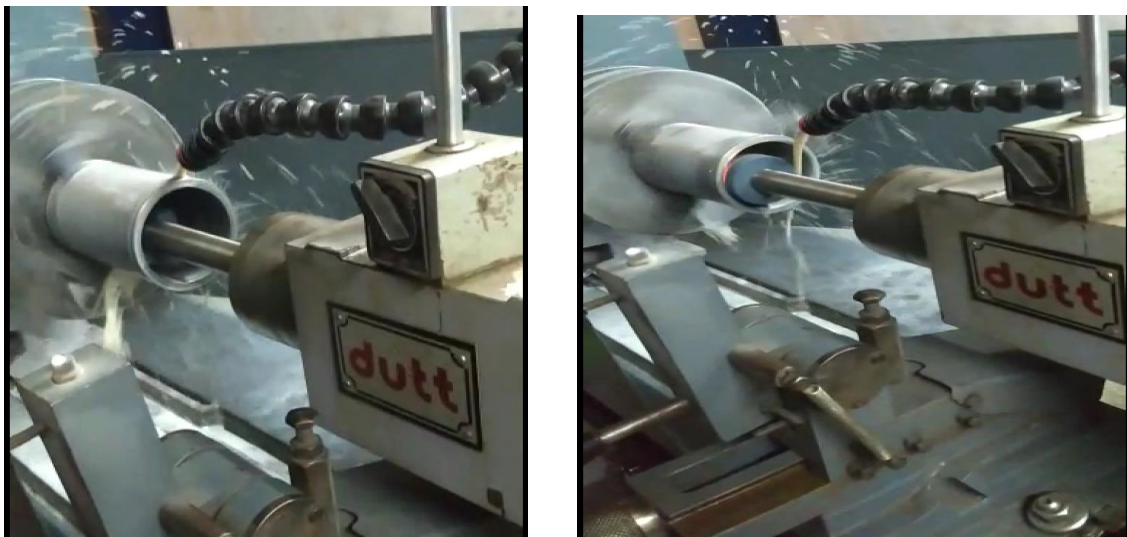
- inserting metal shafts into metal wheels & hubs
- press fitting steel or brass rings around shafts
- fitting guide pins into die shoes
- inserting brass bushings into hubs

Replaces conventional heat and force fitting and eliminates distortion of component parts.

Works faster and more efficiently than dry ice.

Typical shrinkage is:

- 0.002 inches (0.05 mm) per inch diameter on steel
- 0.0035 inches (0.08 mm) per inch on phosphor bronze



Transmission Shaft Sleeve ID grinding Works



Shaft sleeve interference fit 0.05 mm oil heat process (Shaft OD=89.87, Bush ID 89.82)



Transmission Column bush machining work